AIRCRAFT STABILITY AND CONTROL

V Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIA	SEE	Total
A6AE17	PCC	3	-	-	3	40	60	100

UNIT-I — **STATIC LONGITUDINAL STABILITY AND CONTROL**

Introduction Static and dynamic stability-Need for controlled flight - Equilibrium, stability, control, trim-definitions- examples-Need for stability in airplanes. Purpose and types of controls. Static longitudinal stick fixed stability-Stability criterion, Contribution of wing-tail. Effect of fuselage and nacelles, Power effects, Elevator power, Elevator to trim,. Stick fixed neutral point

UNIT-II — **STATIC LONGITUDINAL STABILITY - CONTROL FREE AND MANEUVER STABILITY**

Stick free Static longitudinal stability, Hinge moment coefficients, Control forces to trim. Control free neutral point - Trim tabs. Aerodynamic balancing of control surfaces. Control deflections and control forces in coordinated turns. Control deflection and force gradients. Control fixed and control free maneuver stability-Maneuver points. Maneuver margins

UNIT-III — **INTRODUCTION- EQUATIONS OF MOTION**

Airplane coordinate system-Degree of freedom of a system-Equations of motion of a rigid body-position and orientation of airplane-gravitational and thrust forces-small disturbance theory linearization of equations of motion-aerodynamic forces and moment representation

UNIT-IV — **AERODYNAMIC STABILITY DERIVATIVES**

Aerodynamic stability and control derivatives-derivative due to change in forward speed-derivatives due to pitching velocity2derivatives due to time change of angle of attack-lateral stability derivatives- Derivatives of side force, rolling and yawing moments with respect to the angle of sideslip, rate of sideslip

UNIT-V — **STATIC LATERAL AND DIRECTIONAL STABILITY AND CONTROL,DYNAMIC**

Lateral and directional stability- definition-Dihedral effect, Coupling between rolling and yawing moment, Adverse yaw, Aileron power, Aileron reversal. Weather cocking effects, Rudder power. Control surface deflections in steady sideslips, rolls and turns one engine inoperative conditions, Rudder lock. The principal modes,Phugoid, Short Period Dutch Roll and Spiral modes, Auto rotation and spin recovery

Text Books:

1. Perkins C. D, Robert Hage E (2003), Airplane Performance, Stability and Control, Wiley Toppan, USA.
2. Nelson R. C (2007), Flight Stability and Automatic Control, SIE edition, McGraw Hill, New York.

Reference Books:

1. T. R. Yechout, S. L. Morns (2003), Introduction to Aircraft Flight Mechanics, AIAA Publishers, USA
2. Mc. Cormic B. W. (2010), Aerodynamics, Aeronautics and Flight Mechanics, Wiley India Pvt. Ltd, USA

Course Outcomes : Students should be able to:

At the end of the course the students are able to

1.Formulate the contribution of different components for the static longitudinal stability of the aircraft longitudinal stability
2.Distinguish between static longitudinal stick fixed and stick free stability of the aircraft
3.Formulate the rigid body equations of motion of an airplane
4.Develop the expressions for the aerodynamic and stability derivatives of the aircraft
5.Illustrate the longitudinal, lateral -directional motions of the aircraft in dynamics